GATE ME 90 Days Challenge

Day 32/90 · Practice Set with Detailed Solutions

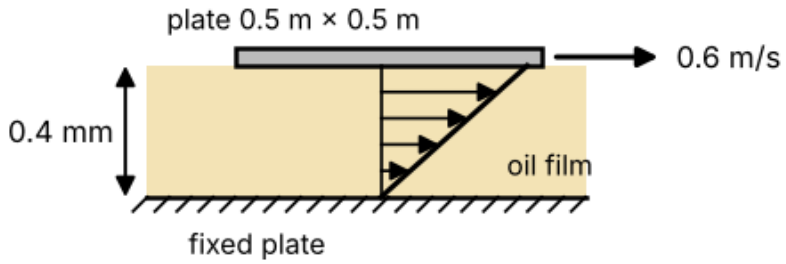
SUBJECT	TODAY'S TOPIC	COUNTDOWN
Fluid Mechanics	Viscous Flow & Boundary Layer	74 days to GATE

How to use this set: 15 questions (8 MCQ, 7 NAT) on viscous flow between plates and in pipes (Couette and Poiseuille flow, Hagen–Poiseuille), bearings, Stokes' law, and laminar boundary layers (Blasius, displacement and momentum thickness, drag, separation). Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.
Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

A 0.5 m × 0.5 m plate is pulled at 0.6 m/s over a 0.4 mm thick oil film resting on a fixed surface, as shown. The force required is 15 N. Assuming a linear velocity profile, the dynamic viscosity of the oil is



(A) 0.04 Pa·s
(B) 0.4 Pa·s
(C) 0.004 Pa·s
(D) 0.024 Pa·s

Q2 NAT

Oil (dynamic viscosity 0.5 Pa·s) flows in laminar regime between two fixed parallel plates 10 mm apart. The maximum velocity is 0.6 m/s. The shear stress at the plates (in Pa) is _____

Q3 MCQ

Oil flows in laminar regime through a pipe under a fixed pressure difference between its ends. If the pipe diameter is increased by 10 %, the discharge increases by about

(A) 10 %
(B) 21 %
(C) 33.1 %
(D) 46.4 %

Q4 NAT

Oil (density 850 kg/m³, dynamic viscosity 0.1 Pa·s) is pumped at 8 litres/s through a horizontal pipe of 100 mm diameter and 200 m length. Take g = 9.81 m/s². The head loss due to friction (in m, round off to two decimal places) is _____

Q5 NAT

A vertical shaft of 200 mm diameter rotates at 400 rpm in a footstep bearing. The end of the shaft is separated from the bearing surface by an oil film 1 mm thick (dynamic viscosity 0.15 Pa·s). The power absorbed in overcoming viscous resistance (in W, round off to one decimal place) is _____

Q6 MCQ

Which statement is **NOT** correct?

(A) Flow in a circular pipe is usually laminar for Re (based on diameter) below about 2000

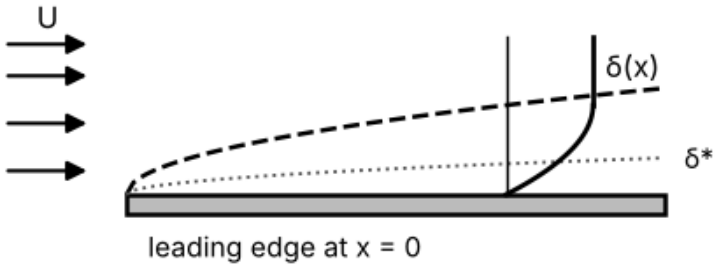
(B) The boundary layer on a smooth flat plate usually becomes turbulent at Re_x of about 5×10^5

(C) In fully developed laminar pipe flow, the Darcy friction factor depends on the pipe roughness

(D) In fully developed pipe flow the velocity profile no longer changes along the pipe

Q7 MCQ

For the boundary layer on a flat plate shown, with δ = boundary-layer thickness, δ* = displacement thickness and θ = momentum thickness, the correct relation is



(A) δ > θ > δ*

(B) δ* > δ > θ

(C) δ > δ* > θ

(D) θ > δ* > δ

Q8 NAT

The velocity profile in a laminar boundary layer is approximated by $u/U = 2(y/\delta) - (y/\delta)^2$. The shape factor $H = \delta^*/\theta$ (round off to one decimal place) is _____

Q9 NAT

Air (kinematic viscosity 1.5×10^{-5} m²/s) flows at 2 m/s over a smooth flat plate. Using $\delta = 5x/\sqrt{Re_x}$, the distance from the leading edge at which the laminar boundary layer is 5 mm thick (in mm, round off to the nearest integer) is _____

Q10 MCQ

A laminar boundary layer exists over the whole length of a flat plate. If the free-stream velocity is doubled (the boundary layer stays laminar), the skin-friction drag becomes

(A) 2 times

(B) 2.83 times

(C) 4 times

(D) 1.41 times

Q11 MCQ

Which of the following does **NOT** help to delay boundary-layer separation?

(A) Suction of the slow fluid near the wall through slots

(B) Making the boundary layer turbulent with a trip wire or vortex generators

(C) Increasing the adverse pressure gradient by a sharper divergence of the passage

(D) Streamlining the body so that the pressure rises gently

Q12 NAT

A sand grain of 0.1 mm diameter (density 2650 kg/m³) settles in still water (density 1000 kg/m³, dynamic viscosity 10^{-3} Pa·s). Take g = 9.81 m/s². Using Stokes' law, its terminal settling velocity (in mm/s, round off to two decimal places) is _____

Q13 MCQ

In fully developed laminar flow through a pipe, the flow rate is doubled while the flow remains laminar. The Darcy friction factor f and the head loss h_f become, respectively,

(A) half and double

(B) double and double

(C) half and four times

(D) unchanged and double

Q14 MCQ

Using the von Kármán momentum-integral equation with the linear profile u/U = y/δ for a laminar boundary layer on a flat plate, the boundary-layer thickness is

(A) $\delta/x = 5.0/\sqrt{Re_x}$

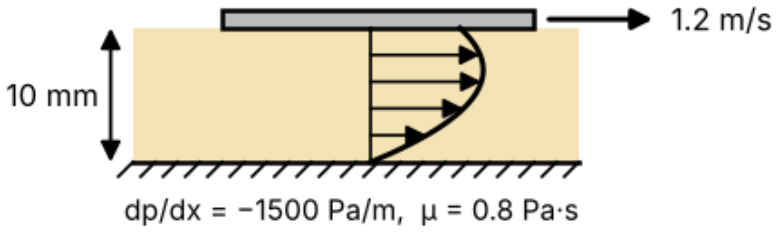
(B) $\delta/x = 3.46/\sqrt{Re_x}$

(C) $\delta/x = 5.48/\sqrt{Re_x}$

(D) $\delta/x = 4.64/\sqrt{Re_x}$

Q15 NAT

Oil (dynamic viscosity 0.8 Pa·s) flows between a fixed lower plate and an upper plate moving at 1.2 m/s, 10 mm apart, with a pressure gradient dp/dx = −1500 Pa/m in the direction of motion, as shown. The shear stress on the fixed plate (in Pa, round off to one decimal place) is _____



Answer Key

Q	Type	Answer
Q1	MCQ	(A) 0.04 Pa·s
Q2	NAT	120
Q3	MCQ	(D) 46.4 %
Q4	NAT	7.82 (accept 7.75 to 7.88)
Q5	NAT	41.3 (accept 40.9 to 41.8)
Q6	MCQ	(C) In fully developed laminar pipe flow, the Darcy friction factor depends on the pipe roughness
Q7	MCQ	(C) $\delta > \delta^* > \theta$
Q8	NAT	2.5
Q9	NAT	133 (accept 132 to 134)
Q10	MCQ	(B) 2.83 times
Q11	MCQ	(C) Increasing the adverse pressure gradient by a sharper divergence of the passage
Q12	NAT	8.99 (accept 8.90 to 9.08)
Q13	MCQ	(A) half and double
Q14	MCQ	(B) $\delta/x = 3.46/\sqrt{Re_x}$
Q15	NAT	103.5

Detailed Solutions

Q1 **Answer: (A) 0.04 Pa·s**

Concept: Newton's law of viscosity for a thin film with a linear (Couette) profile: $\tau = \mu U/h$.

$\tau = F/A = 15/(0.5 \times 0.5) = 60$ Pa

$\mu = \tau h/U = 60 \times 0.0004/0.6 =$ **0.04 Pa·s**

Exam tip: Convert the film thickness to metres first; a missed 10^{-3} gives the distractor 40 or 0.04×10^3.

Q2 **Answer: 120**

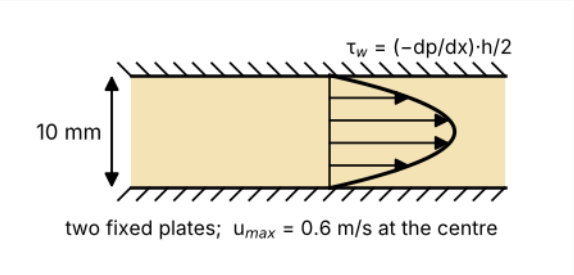
Concept: Plane Poiseuille flow: $u_{max} = (-dp/dx)h^2/(8\mu)$ and $\tau_w = (-dp/dx)h/2$.

$-dp/dx = 8\mu u_{max}/h^2 = 8 \times 0.5 \times 0.6/(0.01)^2 = 24\,000$ Pa/m

$\tau_w = 24\,000 \times 0.01/2 =$ **120 Pa**

Short cut: $\tau_w = \mu(du/dy)_{wall} = 4\mu u_{max}/h = 4 \times 0.5 \times 0.6/0.01 = 120$ Pa.

Exam tip: Between plates $\bar{u} = (2/3)u_{max}$; in a pipe $\bar{u} = u_{max}/2$. Do not mix them up.



Q3 **Answer: (D) 46.4 %**

Concept: Hagen–Poiseuille: $Q = \pi\Delta p D^4/(128\mu L)$, so $Q \propto D^4$ at a fixed Δp.

$Q_2/Q_1 = 1.1^4 = 1.4641 \Longrightarrow$ increase $\approx$ **46.4 %**

Exam tip: At a fixed Q the head loss scales as $1/D^4$; at a fixed Δp the flow scales as D^4. 21 % is 1.1^2, 33.1 % is 1.1^3.

Q4 **Answer: 7.82 (accept 7.75 to 7.88)**

Concept: Check the regime first; if laminar, $h_f = 32\mu VL/(\rho g D^2)$.

$V = 0.008/(\pi/4 \times 0.1^2) = 1.0186$ m/s; $Re = 850 \times 1.0186 \times 0.1/0.1 = 866 < 2000 \rightarrow$ laminar

$\Delta p = 32 \times 0.1 \times 1.0186 \times 200/0.1^2 = 65190$ Pa; $h_f = \Delta p/(\rho g) = 65190/(850 \times 9.81) =$ **7.82 m**

Pumping power $= \Delta p Q = 522$ W.

Exam tip: The same answer comes from Darcy–Weisbach with $f = 64/Re$; use whichever is faster.

Q5 **Answer: 41.3 (accept 40.9 to 41.8)**

Concept: At radius r the film velocity is ωr, so $\tau = \mu\omega r/t$. Torque on a ring: $dT = \tau(2\pi r\, dr)r$.

$T = \int_0^R (\mu\omega r/t)(2\pi r^2)\, dr = \pi\mu\omega R^4/(2t)$

$\omega = 2\pi \times 400/60 = 41.8879$ rad/s; $T = \pi \times 0.15 \times 41.8879 \times 0.1^4/(2 \times 0.001) = 0.9870$ N·m

$P = T\omega =$ **41.3 W**

Exam tip: In a journal bearing every point of the film moves at the same speed, so $T = \tau(\pi DL)(D/2)$ directly; a footstep or collar bearing needs the integral because the speed varies with r.

Q6 **Answer: (C) In fully developed laminar pipe flow, the Darcy friction factor depends on the pipe roughness**

Concept: In laminar flow the viscous effects dominate and the roughness elements are buried in the slow-moving fluid.

Laminar pipe flow: $f = 64/Re$, independent of roughness. So (C) is the incorrect statement.

Exam tip: Roughness matters only in turbulent flow (transitional and fully rough zones of the Moody chart).

Q7 **Answer: (C) $\delta > \delta^* > \theta$**

Concept: $\delta^* = \int(1 - u/U)dy$ and $\theta = \int(u/U)(1 - u/U)dy$; since $0 \le u/U \le 1$, the integrand of θ is smaller than that of δ^*, and both are smaller than δ.

Hence **$\delta > \delta^* > \theta$**, and the shape factor $H = \delta^*/\theta > 1$ (2.59 for Blasius laminar flow, about 1.3 to 1.4 for turbulent flow).

Exam tip: δ^* is the distance by which the wall would have to be displaced outward to carry the same flow in an ideal (inviscid) stream.

Q8 **Answer: 2.5**

Concept: Integrate with $\eta = y/\delta$ from 0 to 1.

$\delta^*/\delta = \int(1 - 2\eta + \eta^2)d\eta = 1 - 1 + 1/3 = 1/3$

$\theta/\delta = \int(2\eta - \eta^2)(1 - 2\eta + \eta^2)d\eta = 2/15$

$H = (1/3)/(2/15) = \mathbf{2.5}$

Exam tip: Linear profile: H = 3; parabolic: 2.5; Blasius exact: 2.59.

Q9 **Answer: 133 (accept 132 to 134)**

Concept: $\delta = 5x/\sqrt{(Ux/\nu)} = 5\sqrt{(\nu x/U)}$, so $x = U\delta^2/(25\nu)$.

$x = 2 \times (0.005)^2/(25 \times 1.5 \times 10^{-5}) = 0.1333$ m = **133 mm**

Check: $Re_x = 17778 < 5 \times 10^5$, so the laminar formula applies.

Exam tip: Since $\delta \propto \sqrt{x}$, doubling the distance multiplies δ by $\sqrt{2}$ only.

Q10 **Answer: (B) 2.83 times**

Concept: $F_D = C_D(\frac{1}{2}\rho U^2)A$ with $C_D = 1.328/\sqrt{Re_L} \propto U^{-1/2}$.

So $F_D \propto U^{3/2}$; doubling U gives $2^{1.5} =$ **2.83 times**.

Exam tip: In laminar flow $F_D \propto U^{1.5}L^{0.5}$; in turbulent flow ($C_D \propto Re^{-1/5}$) $F_D \propto U^{1.8}L^{0.8}$.

Q11 **Answer: (C) Increasing the adverse pressure gradient by a sharper divergence of the passage**

Concept: Separation happens when the near-wall fluid, already slowed by friction, cannot climb an adverse pressure gradient.

Suction removes the slow fluid; a turbulent layer carries more momentum near the wall; streamlining keeps dp/dx small. A sharper divergence makes dp/dx more adverse and **promotes** separation, so (C).

Exam tip: This is why diffuser half-angles are kept small (about 3° to 4°) and why golf balls have dimples.

Q12 **Answer: 8.99 (accept 8.90 to 9.08)**

Concept: At terminal velocity, weight − buoyancy = Stokes drag $3\pi\mu Vd \Longrightarrow V = (\rho_s - \rho)gd^2/(18\mu)$.

$V = 1650 \times 9.81 \times (10^{-4})^2/(18 \times 10^{-3}) = 0.00899$ m/s = **8.99 mm/s**

Check: $Re = \rho Vd/\mu = 0.90 < 1$, so Stokes' law is valid.

Exam tip: Stokes drag means $C_D = 24/Re$; always check $Re < 1$ before trusting it.

Q13 **Answer: (A) half and double**

Concept: $f = 64/Re \propto 1/V$, and $h_f = fLV^2/(2gD) \propto (1/V)V^2 = V$.

Doubling Q doubles V: f is **halved** and h_f is **doubled**.

Exam tip: Laminar $h_f \propto V$; fully rough turbulent $h_f \propto V^2$.

Q14 **Answer: (B) $\delta/x = 3.46/\sqrt{Re_x}$**

Concept: $\tau_w = \rho U^2 \, d\theta/dx$ with $\tau_w = \mu U/\delta$ and $\theta = \delta/6$.

$\mu U/\delta = \rho U^2 (1/6) d\delta/dx \Longrightarrow \delta \, d\delta = 6\nu \, dx/U \Longrightarrow \delta^2 = 12\nu x/U$

$\delta/x = \sqrt{12}/\sqrt{Re_x} = \mathbf{3.46/\sqrt{Re_x}}$

Exam tip: Parabolic profile gives 5.48, cubic gives 4.64, Blasius exact gives 5.0 (4.91 to 99 %).

Q15 **Answer: 103.5**

Concept: Couette–Poiseuille flow: $u = Uy/h + (1/(2\mu))(-dp/dx)(hy - y^2)$.

du/dy at y = 0: $U/h + (-dp/dx)h/(2\mu)$

$\tau_0 = \mu U/h + (-dp/dx)h/2 = 0.8 \times 1.2/0.01 + 1500 \times 0.01/2 = 96 + 7.5 = \mathbf{103.5\ Pa}$

Exam tip: A favourable gradient adds to the Couette shear on the fixed plate; an adverse one subtracts and can even reverse the flow near that plate.

What next?

Day 32 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

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Next, Day 33/90: General Aptitude, Analytical Reasoning. Comment **GATE90** on our reel to get all links. Want concise revision notes? The GATE ME handwritten short notes of an AIR 16 topper are available in the Dig Career Thrust app.